

Database Migration Notes

Hall Pass — Auto-End Forgotten Passes

PulseEDU · School Operations App

Date: July 7, 2026

Scope: schema + data changes shipped today

Commit: 7c37a97 — "Improve hall pass system by automatically ending forgotten passes"

1. Summary

Today's work added a school-configurable "auto-end" threshold for hall passes so that a pass a student forgets to close is automatically ended after a set number of minutes. This required exactly ONE additive database change plus one new value written to an existing text column. There are no destructive changes, no drops, no renames, and no data backfill.

DB objects touched

- Table: school_settings — one new column added.
- Table: hall_passes — no structural change; the existing status column now also stores a new value "auto_ended".

2. Schema change — new column

A new column hall_pass_auto_end_minutes is added to school_settings. It is NOT NULL with a server default of 20 (minutes), so existing rows are backfilled with 20 automatically by the default.

Column definition

Table: school_settings · Column: hall_pass_auto_end_minutes · Type: integer · NOT NULL · DEFAULT 20

Forward migration (idempotent, additive)

```
ALTER TABLE school_settings
  ADD COLUMN IF NOT EXISTS hall_pass_auto_end_minutes
  INTEGER NOT NULL DEFAULT 20;
```

This is applied automatically on API-server boot (see seed.ts ensure-block), matching the project convention of additive "ALTER TABLE ... IF NOT EXISTS" statements rather than interactive drizzle-kit migrations. Running it by hand is safe and repeatable.

ORM (Drizzle) definition

lib/db/src/schema/schoolSettings.ts

```
hallPassAutoEndMinutes: integer('hall_pass_auto_end_minutes')
  .notNull()
  .default(20),
```

Rollback (only if you must fully revert)

```
ALTER TABLE school_settings
  DROP COLUMN IF EXISTS hall_pass_auto_end_minutes;
```

Note: dropping the column discards each school's configured threshold. Not required for a normal deploy — the column is additive and backward compatible.

3. Data / semantic change — new status value

The `hall_passes.status` column is a plain text column (there is no Postgres enum type to alter). A new value, `"auto_ended"`, is now written by the application to passes that exceed the configured threshold while still active. No migration statement is needed for this, but downstream consumers must be aware of it.

Behavior

- When a pass is still "active" past N minutes (N = the school's `hall_pass_auto_end_minutes`, default 20), it is set to `status = "auto_ended"`.
- `ended_at` is capped at `created_at + N minutes` (so a forgotten pass never balloons the recorded duration).
- `ended_by` is set to the literal `"(auto)"`.
- The update is idempotent: it only affects rows still WHERE `status = 'active'`.

Equivalent SQL (for reference / one-off cleanup on a single school)

```
UPDATE hall_passes hp
SET status = 'auto_ended',
    ended_at = to_char((hp.created_at::timestampz + (s.mins * interval '1 minute'))
                      AT TIME ZONE 'UTC', 'YYYY-MM-DD"T"HH24:MI:SS.MS"Z"),
    ended_by = '(auto)'
FROM (SELECT school_id, hall_pass_auto_end_minutes AS mins
      FROM school_settings) s
WHERE hp.school_id = s.school_id
      AND hp.status = 'active'
      AND hp.created_at::timestampz < now() - (s.mins * interval '1 minute');
```

The application performs the equivalent per-row logic in `lib/hallPassLifecycle.ts` (`autoEndStalePasses`). You do NOT need to run this SQL — it is included only so a developer can reason about / manually reconcile historical data if desired.

4. Downstream consumers to review

Any query or report that filters, groups, or displays hall-pass status must account for the new `"auto_ended"` value alongside the existing `"active"`, `"ended"`, and `"system_ended"`.

- Status filters: treat `"auto_ended"` as a terminal (closed) state, like `"ended"/"system_ended"` — not `"active"`.
- Duration/analytics: auto-ended passes have an `ended_at`, so duration math works, but the value is capped at the threshold.
- UI already maps `"auto_ended"` -> label `"Auto Ended"` (styled like an overdue/attention badge).
- Every server surface that reads `"active"` passes now first runs `autoEndStalePasses(schoolId)`: staff pass log, kiosk queue, teacher companion queue, and kiosk pass creation.

5. Deploy checklist

1. Deploy code — the boot ensure-block runs the ALTER automatically; no manual step required.
2. (Optional) Verify the column exists: see verification query below.
3. No data backfill needed — DEFAULT 20 populates existing rows.
4. No downtime, no locks of concern (ADD COLUMN with a constant default is fast in modern Postgres).

Verification query

```
SELECT column_name, data_type, column_default, is_nullable
FROM information_schema.columns
WHERE table_name = 'school_settings'
      AND column_name = 'hall_pass_auto_end_minutes';
-- expect: integer | 20 | NO
```

on this date.